Statistical analysis of the risk factors of chronic respiratory diseases

Statistical Learning 2025 - Final Project

at the
Faculty of Economics and Business

at the
Université de Neuchâtel

with
Prof. Dr. Catalin Starica

as part of the
Bachelor of Science in Data Science
&
Bachelor of Science in Economics

Authors and student numbers:

Nilay Temel / 21-889-662
Verica Dimitrova / 22-826-721
Mariia Ponomarova / 21-507-710
Laila Ibrahim / 23-504-897
Salsabil Mtiraoui / 20-506-630
Kamila Abazi / 23-507-155
Clara Comanescu / 23-507-403

Table of Content

1 Introduction **4**

2 Data **4**

2.1 Data gathering 4

2.2 Descriptive overview 4

3 Methodology **5**

4 Analysis **6**

4.1 Analysis 6

5 Conclusion **7**

6 Bibliography **8**

A Appendix - Supplements **9**

B Appendix - R Code **13**

List of Figures

1 Overview of the variables 9

2 Output of the lm function 10

3 Regression Line for PM25 with Smoking Constant 10

4 Regression Line for Smoking with PM25 Constant 11

5 Residuals vs. fitted values 11

6 Residuals' qq plot 12

7 Studentized Breusch-Pagan test 12

Executive Summary

This project examines the relationship between chronic respiratory diseases (CRD) and two potential risk factors: air pollution and smoking. Using data from 134 countries, we analyzed the impact of fine particulate matter (PM2.5) and smoking rates on CRD prevalence.

As part of this study, we formulated two main hypotheses:

- Higher levels of air pollution (PM2.5) are associated with an increase in chronic respiratory diseases.
- Higher smoking rates are also associated with an increase in chronic respiratory diseases.

Three separate datasets on CRD rates, PM2.5 concentrations and smoking rates were merged using the merge() function in R, followed by cleaning with na.omit() to retain only complete observations. We then conducted an exploratory data analysis to better understand the distribution of the variables, and applied a multiple linear regression model to assess the relationships.

The results show that higher levels of PM2.5 are significantly associated with an increase in CRD rates. On the other hand, smoking does not appear to have a statistically significant effect when air pollution is taken into account. The model explains a moderate part of the variation in CRD, but the reliability of the results is limited by the presence of heteroscedasticity and outliers. These limitations are discussed, and we suggest that including additional relevant variables or using more robust statistical methods could improve the analysis.

1 Introduction

Chronic respiratory diseases such as asthma and chronic bronchitis are affecting more and more people around the world. These conditions can make daily life difficult and often require long-term medical care. Several factors may contribute to their development or worsening, particularly the quality of the air we breathe and certain habits such as smoking.

In this project, we aim to study two main relationships : on one hand, the link between air quality and chronic respiratory diseases, and on the other hand, the link between smoking rates and these same diseases. We will seek to answer the following questions :

- Does air pollution increase the risk of developing respiratory diseases ?
- Is smoking associated with a higher prevalence of these diseases ?

By analyzing these relationships, our goal is to better understand how environmental factors and certain behaviors are associated with the development or worsening of chronic respiratory diseases.

2 Data

We decided to focus our study on the relationship between the quantitative variable "Chronic diseases" and these quantitative variables: PM2.5 and smoking. The study covers 90 countries worldwide. To start with, we will define each of our variables:

- Chronic diseases (CRD) : Rate of chronic respiratory diseases per country, measured as the number of cases per 100'000 inhabitants.
- PM2.5: Average annual air pollution level of fine particles ($\leq$ 2.5 (µm)) in the air, measured in (µg/m^3).
- Smoking: Smoking rate in each country, expressed as a % of the adult population.

2.1 Data gathering

The data was collected from two sources: "chronic diseases"Our World in Data (2024) and "smoking"Our World in Data (2015) tables from ourworldindata.org, while the "PM2.5" data, from iqair.com IQAir (2023).
All three datasets covered the same group of 90 countries, allowing us to merge them into a single table. We selected "chronic diseases" as the analysis variable (y), and "PM2.5" and "smoking" as the explanatory variables (x1 and x2). The final combined dataset was then imported into R software to begin our statistical analysis.

2.2 Descriptive overview

Figure 1 shows us our three variables for the year 2019 across different countries. Firstly, around 80% of countries have a mortality rate due to chronic respiratory diseases of less than 50 deaths per 100,000 inhabitants, while some extreme cases exceed 150 deaths, observed by the outliers in the boxplot. The distribution is asymmetrical, concentrated on low values, as we can see from the QQ plot, which deviates from normality.

For air pollution, most countries have concentrations between 10 and 30 (µg/m^3), with an average concentration around 20 (µg/m^3). The distribution is again pulled to the right by extreme values above 50 (µg/m^3). According to the QQ plot, we do not observe a normal distribution.
Finally, the smoking rate shows a more symmetrical distribution, with around 50% of countries between 15% and 30%. The distribution in the QQ plot is this time close to the theoretical straight line, highlighting a near-normal distribution.

3 Methodology

The analysis was conducted in three main stages, each focusing on the study of the relationship between chronic respiratory disease (CRD) and the two potential explanatory variables: air pollution (PM2.5) and smoking prevalence.

The first stage involved data preparation and cleaning. Three separate datasets - on CRD rates, PM2.5 concentrations and smoking rates - were merged using the merge() function in R. Due to technical constraints associated with merging multiple tables and the presence of missing values, we opted for a two-stage merge, followed by cleaning with na.omit() to retain only complete observations. The final dataset includes 90 countries.

The second stage was devoted to exploratory data analysis (EDA). We calculated descriptive statistics (mean, median, standard deviation, etc.) and visualized the distributions of the three variables using histograms, boxplots and Q-Q plots. These tools enabled us to detect the presence of outliers and assess the normality of the data. We also examined correlations between variables, including after excluding extreme PM2.5 values (values above 60 µg/m^3), to verify their influence on the observed relationships.

The third step involved multiple linear regression modelling, to assess the simultaneous effect of PM2.5 and smoking on CRD rates. The estimated model is of the form :

$$CRD = \beta 0 + \beta 1 \cdot PM2.5 + \beta 2 \cdot Smoking + \epsilon \tag{1}$$

This type of model was chosen for its ability to isolate the specific effect of each explanatory variable, while taking the other into account. Estimation was performed using R's lm() function. Interpretation of the results is based on the p-values of the coefficients, using a significance threshold set at 5% ($\alpha = 0.05$), in accordance with statistical standards. If the p-value was below this threshold, we rejected the null hypothesis (H_0), considering that the variable had a significant effect on CRD.

To ensure the robustness of our model, we carried out several validation tests:

- A plot of residuals versus fitted values to test for homoskedasticity (Figure 5)
- A Q-Q plot of residuals to test normality (Figure 6)
- Breusch-Pagan test (Statology (2021)), using the bptest() function, to detect heteroskedasticity (Figure 7)

Finally, to facilitate interpretation, we produced partial regression plots: holding one of the two explanatory variables constant, we plotted the effect of the other on the DRBs (Figures 3 and 4). These visual representations enhanced understanding of the estimated effects.

By integrating statistical inference, graphical analysis and model diagnostics, we were able to assess the extent to which air pollution and smoking are associated with chronic respiratory disease on an international scale. The limitations of this approach, notably the presence of outliers and heteroskedasticity, are discussed in the following sections.

4 Analysis

4.1 Analysis

In this analysis, we used a linear regression model to investigate the relationship between chronic respiratory disease (CRD) and our two explanatory variables: PM2.5 and Smoking. This allowed us to assess how each factor contributes to CRD when the other is held constant.

The intercept of the model is estimated at 7.0893 with a p-value of 0.38, indicating that it is not statistically significant. Since it represents the expected CRD value when both PM2.5 and Smoking are zero, an unrealistic scenario, this value does not offer meaningful interpretation in practice. Therefore, we focus on the coefficients of the explanatory variables. The coefficient for PM2.5 is 1.0391, with a highly significant p-value of 4.5e-07. This indicates a strong positive relationship: for every unit increase in PM2.5, CRD increases by approximately 1.04 units, assuming Smoking remains constant. On the other hand, the coefficient for Smoking is 0.2043, with a p-value of 0.514. This lack of statistical significance suggests that, within this model, Smoking does not have a measurable effect on CRD when controlling for PM2.5. The model's multiple R-squared value is 0.2884, meaning that about 28.8% of the variation in CRD is explained by PM2.5 and Smoking. To account for model complexity, the adjusted R-squared is also reported: 0.2711. This adjusted value provides a more reliable estimate of explanatory power and suggests that roughly 27% of the variability in CRD can be attributed to the included predictors. While this is a moderate value, it also implies that most of the variability in CRD is driven by factors not included in the model. Residual analysis shows that prediction errors range from -51.261 to 120.835, with a median near zero. The centered distribution of residuals suggests unbiased predictions, but the wide spread indicates the presence of outliers and potential model misfit. The residual standard error (SER) is 24.8, meaning that the model's predictions deviate from actual CRD values by an average of 24.8 units. This relatively large error aligns with the moderate adjusted R-squared and suggests limited precision in the predictions.(Figure 2)

Visual inspection supports these findings. The CRD vs PM2.5 scatter plot, holding Smoking constant at its mean (Figure 3), shows a clear positive trend. The upward slope of the regression line reflects the positive association, while the moderate scatter of data points around the line is consistent with the R-squared value. However, several points at higher CRD levels deviate noticeably, suggesting the presence of outliers or influential cases.

In contrast, the CRD versus Smoking plot, with PM2.5 held constant (Figure 4), reveals a nearly flat regression line and a wide dispersion of data points. This visual pattern confirms the lack of significant association between Smoking and CRD found in the regression results. The presence of outliers further weakens any potential signal in this relationship.

A residuals vs fitted values plot (Figure 5) was also examined to assess the assumptions of linear regression. The residuals appear mostly centered around zero, which supports the assumption of unbiased errors. However, the spread of residuals increases with higher fitted values, suggesting

heteroskedasticity—non-constant variance of the error terms. Additionally, some residuals are quite far from zero, indicating the presence of outliers or influential data points. These patterns suggest that the assumption of homoscedasticity is likely violated and that alternative modeling approaches, such as transformations or robust regression techniques, should be considered.

A Q-Q plot (Figure 6) was also used to assess the normality of residuals. Most points lie along the reference line, indicating that the residuals are approximately normally distributed in the central region. However, deviations at the tails suggest that the distribution has heavy tails, which may correspond to the outliers identified earlier. While not a major concern for point estimation, these deviations could affect the accuracy of hypothesis tests and confidence intervals.

To formally test for heteroskedasticity (non-constant variance of residuals), we conducted the studentized Breusch-Pagan test (Figure 7). The test yielded a BP statistic of 14.487 with 2 degrees of freedom and a p-value of 0.000715. Since this p-value is well below the 0.05 threshold, we have strong statistical evidence of heteroskedasticity. This means the variance of the residuals changes across levels of the predicted values, violating a key assumption of linear regression. As a result, standard errors and significance tests may be unreliable unless corrective measures, such as robust standard errors or transformations, are applied.

In summary, the regression model reveals a significant and positive relationship between PM2.5 and CRD, while Smoking appears to have no significant effect when PM2.5 is accounted for. The model explains a moderate portion of the variance in CRD, but the presence of heteroskedasticity and influential outliers limits its reliability. Further analysis using robust methods or additional variables could improve model performance and inference accuracy.

5 Conclusion

This analysis shows that air pollution, measured by PM2.5 levels, is significantly associated with an increased risk of chronic respiratory disease. In contrast, there is not a statistically significant effect of smoking that remains once PM2.5 has been considered. These findings indicate that air pollution is a stronger predictor of CRD in our model and support the conclusion that higher PM2.5 levels is associated with greater CRD prevalence, and that smoking does not appear to have an impact in this context.

Yet, the analysis encounters a few restrictions. The model exhibits heteroskedasticity and includes outliers, both of which can reduce the reliability of the estimates. Moreover, important variables such as age, pre-existing health conditions, or other environmental factors were not taken into consideration which may lead to omitted variable bias. These missing factors could influence both the predictors and the outcome, thus affecting the validity of our results.

To improve the analysis, we could look at more robust regression techniques, applying transformations to address non-constant variance, and incorporating additional relevant variables. These improvements would help provide a more accurate and comprehensive understanding of the factors contributing to CRD.

6 Bibliography

IQAir, "World's Most Polluted Countries in 2023 - PM2.5 Ranking," https://www.iqair.com/world-most-polluted-countries?srsltid=AfmBOopLcWmQQmjRHN_d5HNfaRxPrz9lgCC33AK2QYNpyEQknvij3hy1 2023. Accessed 2023.

Our World in Data, "Share of Adults Who Smoke," https://ourworldindata.org/grapher/share-of-adults-who-smoke?tab=table&time=earliest..2019&country=USA~BRA~RUS~IND~CHN~European+Union~IDN~OWID_WRL 2015. Accessed 2024.

_ , "Chronic respiratory diseases death rate," https://ourworldindata.org/grapher/respiratory-disease-death-rate?tab=table&time=earliest..2019 2024. Accessed 2024.

Statology, "Breusch-Pagan Test: Definition and Example," https://www.statology.org/breusch-pagan-test/ 2021.

A Appendix - Supplements

Figure 1: Overview of the variables

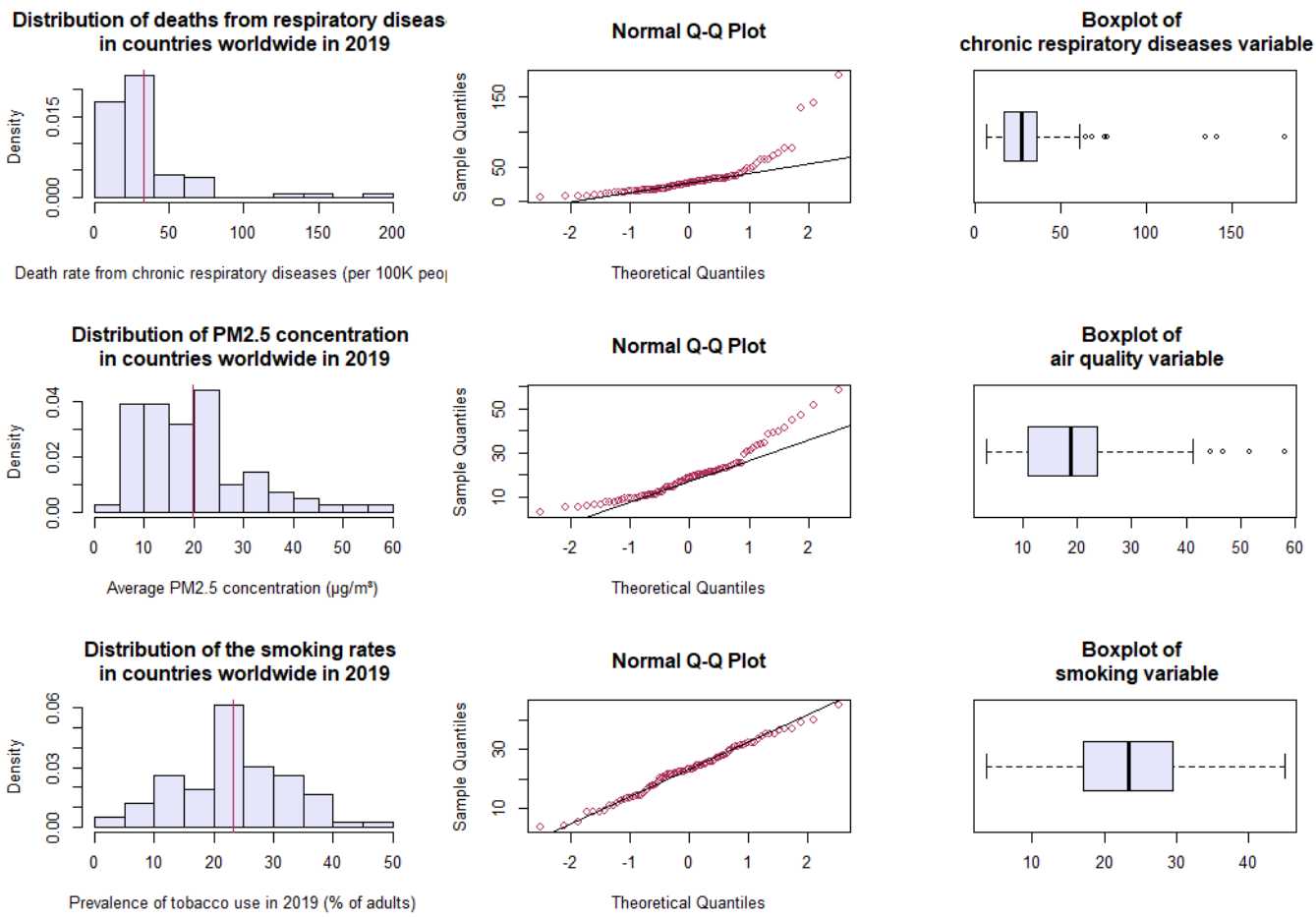

Figure 2: Output of the lm function

```
Call:
lm(formula = crd ~ pm25 + Smoking, data = data)

Residuals:
    Min      1Q  Median      3Q     Max 
-51.261 -14.136  -0.715   7.267 120.835 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept)   7.0893     8.0246   0.883    0.380    
pm25          1.0391     0.1895   5.484  4.5e-07 ***
Smoking       0.2043     0.3118   0.655    0.514    
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 24.8 on 82 degrees of freedom
Multiple R-squared:  0.2884,	Adjusted R-squared:  0.2711 
F-statistic: 16.62 on 2 and 82 DF,  p-value: 8.734e-07
```

Figure 3: Regression Line for PM25 with Smoking Constant

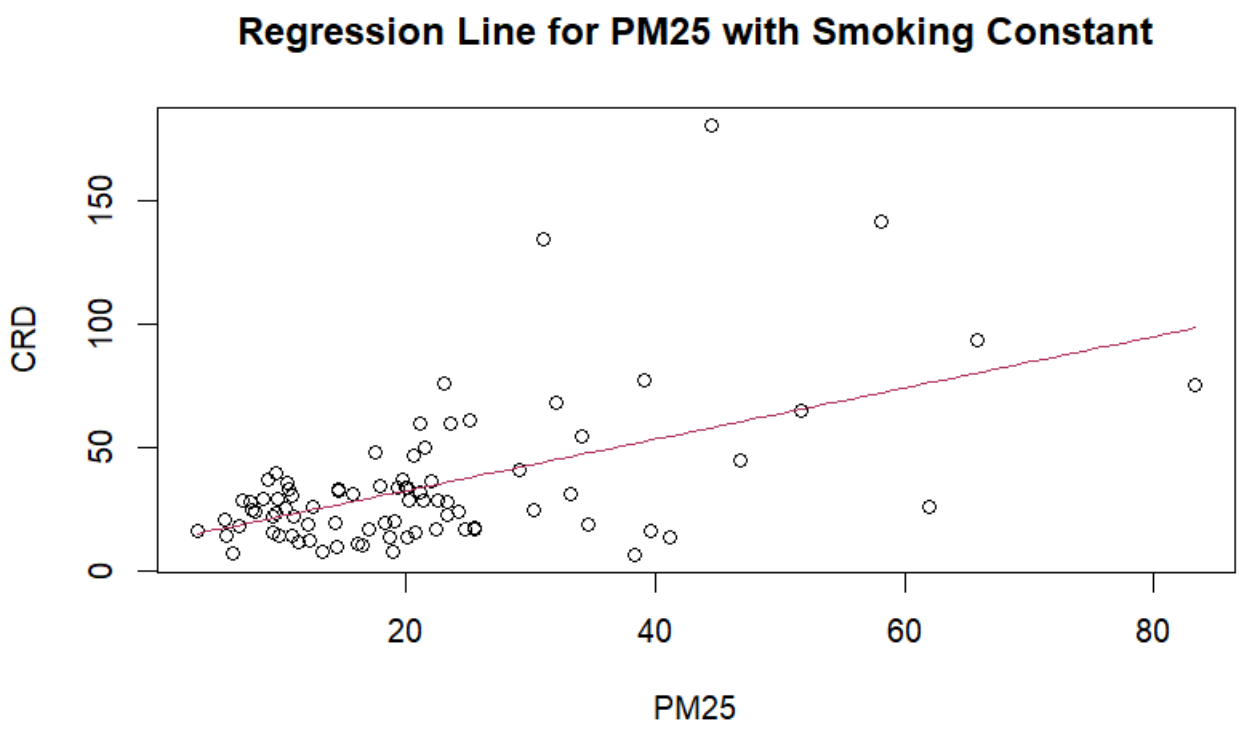

Figure 4: Regression Line for Smoking with PM25 Constant

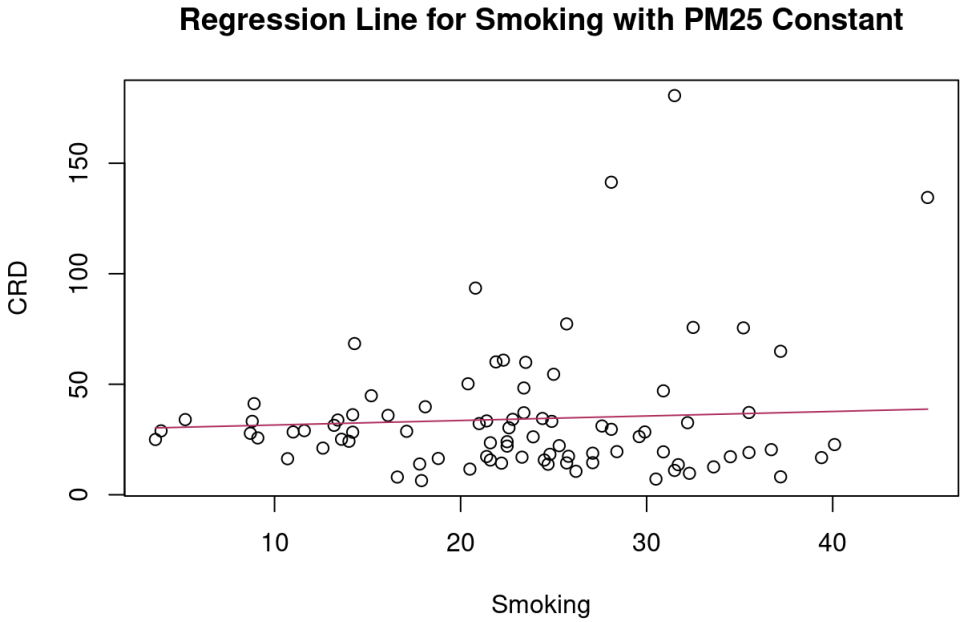

Figure 5: Residuals vs. fitted values

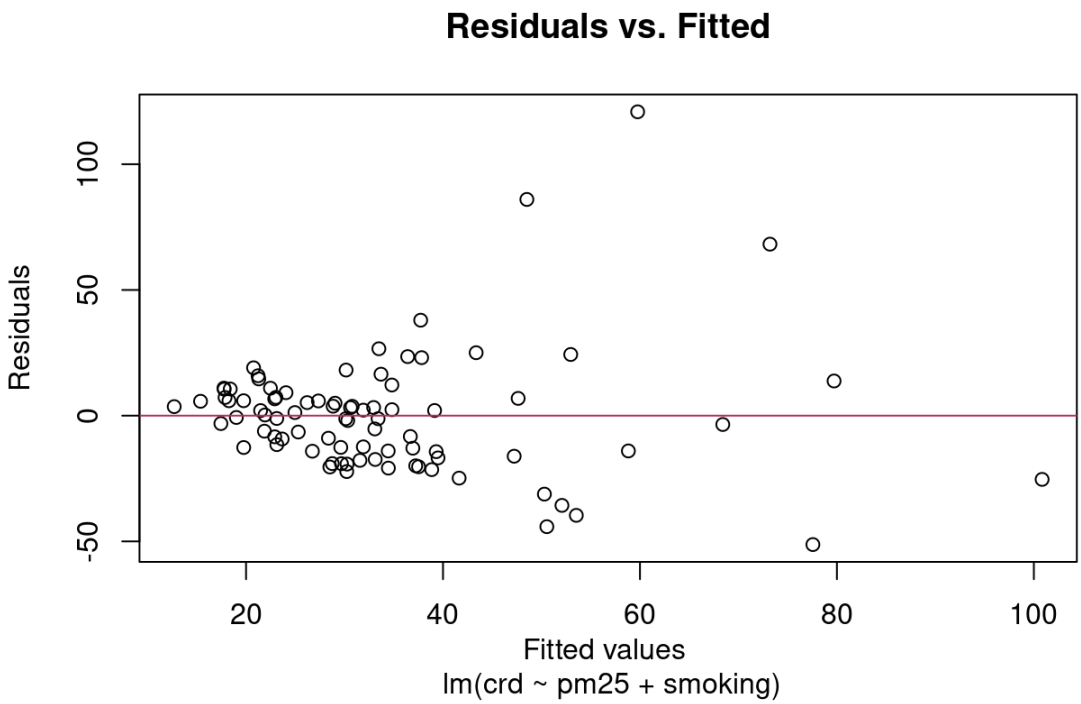

Figure 6: Residuals' qq plot

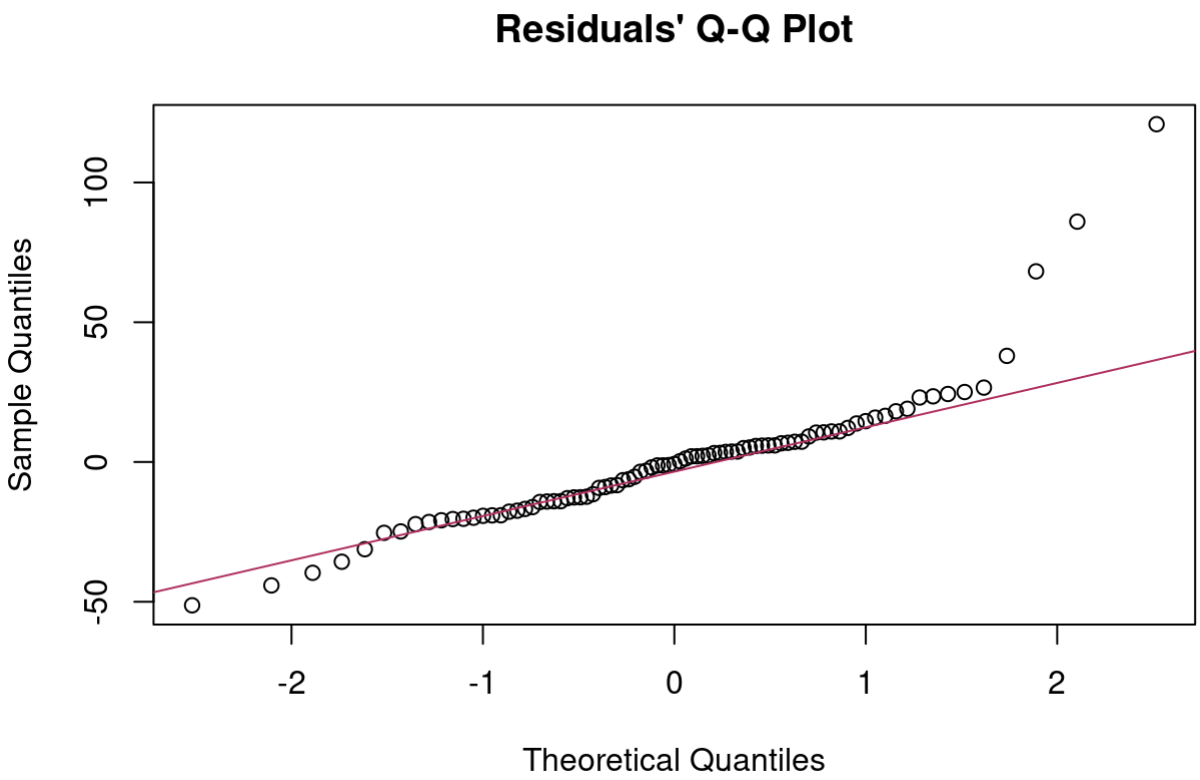

Figure 7: Studentized Breusch-Pagan test

```
        studentized Breusch-Pagan test

data:  mod
BP = 14.487, df = 2, p-value = 0.000715
```

B Appendix - R Code

```
airquality = read.table("PM25.csv", header = TRUE, sep=",")
airquality

resp_dis = read.table("chron-resp.csv", header = TRUE, sep=",")
resp_dis

smoking=read.table("smoking.csv", header = TRUE, sep=",")
smoking

#merging the air quality data and the respiratory diseases rates in one table
data1 = na.omit(merge(airquality, resp_dis, by = "Countries"))
data1

#merging all of the data together
#we had to divide the merging in 2 steps,
#because the merge function could not handle combining
#all of the 3 columns at once
data = na.omit(merge(data1, smoking, by = "Countries"))
data

#setting the variables for easier use
pm25 = data$pm25
crd = data$crd
smoking = data$Smoking

#checking differences between correlations without outliers

subset_indices = which(data$pm25 < 60)
pm25 = data$pm25[subset_indices]
crd = data$crd[subset_indices]
#smoking = data$Smoking[subset_indices]
#smoking
cor(pm25, crd)
#cor(smoking, crd)

#data overview
summary(pm25)
summary(crd)
summary(smoking)


#graphs
par(mfrow=c(3,3))
hist(crd, col = "lavender", xlab = "Death rate from chronic respiratory
   diseases (per 100K people)", main="Distribution of deaths from respiratory
   diseases \n in countries worldwide in 2019", probability = TRUE)
abline(v=mean(crd), col="maroon")
qqnorm(crd, col="maroon")
```

```r
qqline(crd)
boxplot(crd,horizontal = T, main="Boxplot of \n chronic respiratory diseases
   variable", col = "lavender")

hist(pm25, col = "lavender", xlab = "Average PM2.5 concentration ( g / m  )",
   main="Distribution of PM2.5 concentration \n in countries worldwide in
   2019", probability = TRUE)
abline(v=mean(pm25), col="maroon")
qqnorm(pm25, col="maroon")
qqline(pm25)
boxplot(pm25,horizontal = T, main = "Boxplot of \n air quality variable", col
   = "lavender")

hist(smoking, col = "lavender", xlab = "Prevalence of tobacco use in 2019 (%
   of adults)", main="Distribution of the smoking rates \n in countries
   worldwide in 2019", probability = TRUE)
abline(v=mean(smoking), col="maroon")
qqnorm(smoking, col="maroon")
qqline(smoking)
boxplot(smoking,horizontal = T, main = "Boxplot of \n smoking variable", col =
   "lavender")


#fit a linear regression model
mod = lm(crd ~ pm25 + Smoking, data = data)
summary(mod)



#plotting partial effect of PM2.5 on CRD, holding Smoking constant
par(mfrow=c(1,1))
plot(data$pm25, data$crd, xlab = "PM25", ylab = "CRD", main = "Regression Line
   for PM25 with Smoking Constant")
pm25_vals <- seq(min(data$pm25), max(data$pm25), length.out = 100)
mean_Smoking <- mean(data$Smoking, na.rm = TRUE)
predicted_crd <- predict(mod, newdata = data.frame(pm25 = pm25_vals, Smoking =
   mean_Smoking))
lines(pm25_vals, predicted_crd, col = "maroon")



#plotting partial effect of Smoking on CRD, holding PM2.5 constant
plot(data$Smoking, data$crd, xlab = "Smoking", ylab = "CRD", main =
   "Regression Line for Smoking with PM25 Constant")
smoking_vals <- seq(min(data$Smoking), max(data$Smoking), length.out = 100)
mean_pm25 <- mean(data$pm25, na.rm = TRUE)
predicted_crd <- predict(mod, newdata = data.frame(Smoking = smoking_vals,
   pm25 = mean_pm25))
lines(smoking_vals, predicted_crd, col = "maroon")
```

```r
#residuals vs. fitted plot
plot(mod$fitted.values, mod$residuals, xlab = "Fitted values \n lm(crd ~ pm25
   + smoking)", ylab = "Residuals", main = "Residuals vs. Fitted")
abline(h = 0, col = "maroon")

#residuals qq plot
qqnorm(residuals(mod), main = "Residuals' Q-Q Plot")
qqline(residuals(mod), col = "maroon")

#breusch-pagan test
library(lmtest)
bptest(mod)
```

Déclaration d'honneur

Par la présente, j'affirme avoir pris connaissance des documents d'information et de prévention du plagiat émis par l'Université de Neuchâtel et m'être renseigné-e correctement sur les techniques de citation. J'atteste par ailleurs que le travail rendu est le fruit de ma réflexion personnelle et a été rédigé de manière autonome.

Je certifie que toute formulation, idée, recherche, raisonnement, analyse ou autre création empruntée à un tiers est correctement et consciencieusement mentionnée comme telle, de manière claire et transparente, de sorte que la source en soit immédiatement reconnaissable, dans le respect des droits d'auteur et des techniques de citations.

Je suis conscient-e que le fait de ne pas citer une source ou de ne pas la citer clairement, correctement et complètement est constitutif de plagiat. Je prends note que le plagiat est considéré comme une faute grave au sein de l'Université. J'ai pris connaissance des risques de sanctions administratives et disciplinaires encourues en cas de plagiat (pouvant aller jusqu'au renvoi de l'université).

Je suis informé-e qu'en cas de plagiat, le dossier sera automatiquement transmis au rectorat. Au vu de ce qui précède, je déclare sur l'honneur ne pas avoir eu recours au plagiat ou à toute autre forme de fraude.

Neuchâtel, Mai 2025 Les auteur-e-s.

Texte basé sur la ≪Déclaration sur l'honneur≫ du Rectorat de l'Université de Neuchâtel.